

Anonymisation Protocol & Decision Record: OHF Community Survey 2024 - Public Data Release

Controller: Open Home Foundation (CHE-416.988.952), c/o Treforma AG, Grabenstrasse 25, 6340 Baar, Switzerland **Dataset:** Community Survey 2024 (Typeform export, ~8,000 responses, 557 columns) **Destination:** Zenodo (CERN) public repository, open-science release under FAIR principles **Status:** Legal confirmation received; publication approved on completion and documentation of the k-anonymity review **Date:** 20.07.2026

1. Purpose & scope

The Open Home Foundation (OHF) publishes the Community Survey 2024 dataset on Zenodo to support open science and align with the FAIR principles (Findable, Accessible, Interoperable, Reusable). This document records the reasoning and the concrete steps taken to ensure the released dataset is **anonymised** — that is, that no individual can reasonably be re-identified from it — and to align that release with Open Home Foundation's legal and ethical obligations and its privacy-first mission.

The release is assessed against three frameworks:

- The **Swiss Federal Act on Data Protection (nFADP / revFADP, in force 1 Sept 2023)** and its Ordinance (**OFADP**), as Open Home Foundation is a Swiss controller.
- The **EU General Data Protection Regulation (GDPR)**, which applies extraterritorially because the survey pool is global and includes EU/EEA residents.
- The **FAIR principles** for research data publication.

This is a data-governance record capturing the reasoning behind each decision.

2. Legal basis & consent analysis

The survey's lawful basis for collection is **consent** under GDPR Art. 6(1)(a), as stated in the [OHF User Research Agreement](#). That agreement governs what OHF may do with the data.

Once a dataset is **genuinely anonymised**, it is no longer personal data and falls outside the scope of both the nFADP (which protects data relating to identified or identifiable natural persons, nFADP Art. 5(a)) and the GDPR (Recital 26: the principles of data protection do not apply to anonymous information). At that point, public release is not "processing of personal data" and does not require a consent clause covering publication. This is the basis on which the release proceeds, and it is **conditional on anonymisation being complete and defensible**. Every removal decision in §4–§5 exists to make that condition hold.

Two reinforcing constraints:

- **Immutability vs. data-subject rights.** The agreement grants withdrawal of consent (GDPR Art. 7(3)) and erasure (Art. 17) rights, and states that data will be deleted or anonymised once no longer needed. A Zenodo deposit receives a permanent DOI and is downloadable worldwide; it cannot be recalled, and erasure requests cannot be honoured against copies already distributed. If any published row were still personal data, the release would create an un-eraseable store in conflict with these rights. Anonymisation must therefore be **complete before deposit**, not remediated afterward.
- **The non-identification promise.** The agreement's twice-stated promise that participants will not be identified sets the operational bar. The release must be defensible against the question "could any single individual be singled out?", not merely "have direct identifiers been removed?"

3. Standard applied

Pseudonymisation is not sufficient. Removing names while retaining a rich combination of indirect attributes yields pseudonymised data, which remains personal data and fully in scope (GDPR Recital 26; nFADP). The release must clear the higher bar of anonymisation.

Anonymisation test. The dataset is treated as anonymised only where re-identification is not *reasonably likely* by a **motivated intruder** — a person of ordinary competence and resources attempting to single out an individual, including by linking the dataset against external sources (reference: WP29 Opinion 05/2014 on Anonymisation Techniques; the "singling out", "linkability", and "inference" risks).

Quantitative target. For any quasi-identifying attributes retained in the release, a **k-anonymity threshold of $k \geq 10$** is applied: every combination of retained quasi-identifiers must be shared by at least 10 respondents, or be suppressed/generalised until it is. Because all special-category attributes are being removed (§4), l-diversity on sensitive attributes is not separately required.

4. Data classification

Each column in the source export is assigned to one category. Question numbers reference the **survey question map (excel-file)**. "Article" columns cite the primary provision engaged.

Category	What it is	Example questions	GDPR	nFADP	Decision
----------	------------	-------------------	------	-------	----------

Direct identifiers / metadata	Values that identify or trace a respondent or device	Response token (col 0); Network ID / hashed IP (col 554); Start / Stage / Submit timestamps (cols 551–553); source, Response Type, Tags, Ending	Art. 4(1) (personal data); IP as personal data per CJEU <i>Breyer</i> , C-582/14	Art. 5(a)	Drop. Replace token with a fresh sequential ID unlinkable to the original.
Special-category / sensitive data	Health, disability, neurodiversity, religion	Disability & assistive tech (Q5–Q7); neurodiversity & ADHD subtype (Q17–Q19); religion / spirituality (Q73–Q74)	Art. 9	Art. 5(c) (sensitive personal data)	Drop. Highest re-identification stakes and highest harm; public disclosure needs the firmest basis, which is absent.
Quasi-identifiers (demographic)	Attributes that combine to single someone out	Age band (Q15); pronouns (Q16); field of work (Q20); education (Q21); languages at home (Q70–Q72)	Art. 4(1); Recital 26 (singling out)	Art. 5(a)	Drop. Conservative posture; removes the demographic combination axis entirely.
Quasi-identifiers (geographic & dwelling)	Location and home characteristics	Country of primary/secondary home (Q118, Q123); home type / size / floors (Q115–Q117); own vs. rent (Q119)	Art. 4(1); Recital 26	Art. 5(a)	Drop. Geography is the strongest re-identification vector for a global set; the small-country tail is identifying.

Third-party personal data	Data about people who did not take the survey	Partner's pronouns / occupation / education (Q81–Q83); children's ages & count (Q84–Q89); other household members (Q96–Q99); household size (Q75); pets (Q100–Q102)	Art. 4(1); Art. 6 / Art. 9 for third parties	Art. 5(a), 5(c)	Drop. No lawful basis to publish data on non-participating third parties.
Free text	Unbounded open responses	All ~50 "Other" fields; long/short-text questions Q6, Q11, Q30, Q40, Q43, Q59, Q61, Q65, Q72, Q74, Q89, Q96, Q102, Q124, Q130, Q135, Q146, Q154	Art. 4(1)	Art. 5(a)	Drop all. Open text can contain names, employers, locations, and unique circumstances.
Retained — behavioural / technical	Non-identifying setup and usage data	Device categories (Q22–Q29); networking (Q38); ESPHome (Q41–Q44); device count (Q45); protocols (Q46–Q47); HA hardware (Q48–Q49); UI (Q50); HA tenure (Q51); HA motivations (Q52); features (Q56); config style (Q57); historical data (Q58); Cloud usage (Q63–Q64)	—	—	Retain, subject to k-anonymity check (§6). Individually non-identifying; verify no rare combination singles anyone out.
Retained — origin & attitudinal	Platform history and opinion scales	Platform origin (Q8); HA awareness (Q9); onboarding friction (Q10); role (Q12); time spent (Q13);	—	—	Retain, subject to k-anonymity check (§6).

		privacy/sustainability/freedom scales (Q31–Q37)			
--	--	---	--	--	--

Net effect. After removal of direct identifiers, special-category, demographic, geographic, third-party, and free-text columns, the released file contains only coarse behavioural, technical, platform-origin, and attitudinal responses drawn from fixed option sets — the material relevant to general community insight, with the identifying surface removed.

5. Treatment actions

The following steps produce the release file from the source export. Each is scripted so the process is reproducible and auditable.

1. **Load** the raw export and reattach question numbers via the question map (multi-select option columns are otherwise unlabelled).
2. **Remove metadata / trailer columns:** response token, Network ID, all timestamps, source, Response Type, Tags, Ending.
3. **Drop special-category columns** (Q5–Q7, Q17–Q19, Q73–Q74 and their option/"Other" columns).
4. **Drop demographic quasi-identifiers** (Q15, Q16, Q20, Q21, Q70–Q72).
5. **Drop geographic & dwelling quasi-identifiers** (Q115–Q119, Q122–Q125).
6. **Drop third-party / household columns** (Q75, Q77–Q102 where they describe partners, children, housemates, guests, pets).
7. **Drop all free-text columns** — identified programmatically as (a) any column whose header is "Other", plus (b) columns mapped to question types *long text* / *short text* / free-form *dropdown*. Cross-check against the free-text list in §4 rather than name-matching alone.
8. **Assign a new sequential respondent ID** (1...N) with no link table retained, replacing the original token.
9. **Run the k-anonymity check** on all retained columns; generalise or suppress any combination below the threshold.
10. **Export** the anonymised file plus a codebook; record the column-level keep/drop decisions in the audit trail.

Note on the working vs. released dataset. All removals apply to the *published artifact only*. OHF retains the full dataset internally under its existing consent basis and may continue to analyse every field for its own reports. Scoping the public release does not reduce OHF's internal research capacity.

6. FAIR / publication layer

- **Findable:** Zenodo assigns a persistent DOI; complete descriptive metadata (title, creators, description, keywords, publication date).
- **Accessible:** open download via Zenodo; no personal data, so no access restriction required.
- **Interoperable:** release as CSV (UTF-8) with an accompanying machine-readable codebook derived from the question map.
- **Reusable:** include this anonymisation record (or a summary), the codebook, the sampling caveat (survey completion implies engaged users — a non-representative frame), and a clear licence.
- **Licence — CC-BY-4.0:** Two options were considered. **CC0** is a public-domain dedication allowing any reuse with no conditions, and is the common default for anonymised research microdata because it maximises frictionless reuse. **CC-BY-4.0** grants the same reuse freedoms (including commercial and derivative use) on the single condition that reusers credit Open Home Foundation. CC-BY-4.0 is selected to preserve visibility and attribution for Open Home Foundation as a mission-driven foundation. Note that copyright over raw factual data is thin, so the attribution requirement functions primarily as a citation norm rather than a strong enforcement mechanism; this is an accepted trade-off for the visibility benefit.

7. Verification: k-anonymity review & outcome

Before release, the anonymised dataset was tested to confirm that no individual can be singled out from it. The test used **k-anonymity**: for a chosen set of attributes, every combination of values must be shared by at least k respondents. We set **$k = 10$** , meaning any given profile in the released data is indistinguishable from at least nine others. This is a conservative threshold (twice the common baseline of five) chosen to match OHF's privacy-first posture and the fact that the release is public and permanent.

Which attributes were tested. k-anonymity is only meaningful over attributes an outside party could realistically both *know* about a specific person and *use to match them* to a record. We limited the tested set to six such attributes:

1. how long the respondent has used Home Assistant
2. whether their use is private or professional
3. whether they currently use Home Assistant Cloud
4. which platform(s) they used before Home Assistant
5. which communication protocols they use
6. whether they have purchased Home Assistant hardware

These are the kinds of things a person might reveal publicly (for example, in a community forum). Other retained variables are not reasonably observable or linkable from outside.

Generalise first, then suppress. Following standard disclosure-control practice, we first reduced detail so that more respondents shared each profile, and only removed records as a last resort. The generalisation steps were: the three multi-answer attributes were reduced from exact selections to coarse summaries

- prior platforms → none / one / several
- protocols → count bands
- hardware → purchased any HA hardware, yes/no
- cloud use → current-use yes/no
- two very small response categories were merged into adjacent ones
- non-response was treated as its own explicit category so that missing answers do not create artificial uniqueness

After these steps, a residual group of records still fell into profiles shared by fewer than ten people; these were then **suppressed** (removed).

Outcome: Of the 8,619 anonymised records, **1,047 (12.1%) were suppressed**, leaving **7,572 records** in the released dataset. Every profile in the released data is now shared by at least ten respondents (minimum equivalence-class size = 10), so the $k = 10$ standard is met. This completes the verification gate; combined with the anonymisation steps in §5, the release rests on the dataset being genuinely non-identifying.

In plain terms: how we protected user privacy

If you took part in this survey, here is what we did with your answers before sharing any of the data publicly.

Our goal was simple: publish useful survey data for open research **without anyone being able to figure out who any individual respondent is**. Protecting privacy came before everything else, and wherever we faced a choice, we took the more cautious and conservative option, even when that meant keeping less data.

In practice, this meant:

- **We removed anything that could point to a person.** Names, contact details, technical identifiers, timestamps, free-text comments where people sometimes mention personal details, and anything sensitive (such as anything touching on health, disability, religion, or household members) were all taken out. We removed so much identifying information

that the shared dataset is no longer considered "personal data" under European privacy law (GDPR). Legally, it's no longer about identifiable people at all.

- **We verified that no one stands out.** Removing obvious identifiers isn't enough on its own; sometimes a rare *combination* of ordinary details (e.g., a long-time user with an unusual setup) could make someone stand out. So we ran a validated test that looks for exactly this. Where needed, we grouped answers into broader categories: for example, describing someone as using "5–6 protocols" rather than listing the exact protocols. This ensured that each respondent blends in with many others rather than being unique.
- **When in doubt, we cut the data.** After all these steps, a small number of responses still had rare enough combinations that we couldn't be fully confident they'd blend in. Rather than take any chances, we simply removed them from the dataset. In the end we left out about **12% of responses** (roughly one in eight) purely as a precaution. We would rather publish less and be certain than risking any leaks.
- **We checked it legally.** Our approach was iteratively reviewed by legal advisors, and every decision and step is written down in this document so it can be independently examined.

The result is a dataset that lets researchers learn from the community's collective experience, while keeping every individual respondent anonymous. If you have questions about any of this, you're welcome to reach out to us at research@openhomefoundation.org.